

Joséphine Gal

French and Dutch, 30 y. o.

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Aeronautical Project Management student moving from life sciences research into aerospace engineering and design. My research on how organisms adapt to challenging environments introduced me to the diverse strategies they use to overcome physical constraints. Learning about biomimicry led me to explore how nature's solutions inspire innovation in aeronautics and space. I am now seeking a 6-month internship to apply my analytical and problem-solving skills to early-stage structural design, preliminary sizing, and airframe architecture optimisation.

Software & Tools

- 3D modelling and simulation: 3D CAD (FreeCAD, Blender) and finite element analysis
- Programming: R, Python
- Data analysis and visualisation: GraphPad Prism, Excel
- Image automation & object detection: QUPath, Fiji

Investigation & problem-solving skills:

- Experimental design
- Technical troubleshooting
- Process optimisation

Professional skills

- Pragmatic & solution-oriented
- Rigor, attention to detail
- Adaptability & fast learning

Languages

- French: native
- English: advanced
- Swedish: A1

Interests

- Thai boxing
- Mechanical restoration (Renault 4L)
- Hands-on work on the family farm

EDUCATION

2026 – **Advanced Master in Aeronautical and Space Project Management**, Arts et Métiers Institute of Technology, Bordeaux

- **Engineering design and integration:** Structural mechanics, strength of materials, aerodynamics, flight mechanics.
- **Systems engineering:** Requirements definition, model-based systems engineering (MBSE), system integration, human factors, and operational safety and reliability.
- **Hands-on engineering project:** ULM aircraft design project with Airbus, focusing on overall architecture, multidisciplinary integration and engineering trade-offs involving performance, weight, safety, manufacturability and maintainability.
- **Aerospace lifecycle and industrial processes:** Design, ground and flight testing, production, industrial organisation, maintenance and operational support.

2020 – **Engineering degree in Agricultural sciences**, VetAgro Sup, Clermont-Ferrand

PROFESSIONAL EXPERIENCE

Independent research project – **Biomimicry for Aerospace Applications**

Developed a research proposal exploring biological principles with potential applications in space debris capture, robotic grasping, planetary exploration and extraterrestrial sample collection.

Research Scientist – Immune response to tuberculosis

Linköping University, Sweden | 2 years

- Developed and optimised quantitative measurement protocols, improving data reproducibility and robustness.
- Processed and analysed multidimensional datasets using R.

PhD Researcher – Filariasis and pulmonary environment

National Museum of Natural History, Paris | 3 years

- Built automated microscopy workflows for object detection and pathological feature recognition.
- Optimised experimental protocols under strict schedule and budget constraints.